AI/ML Assessment – Problem Statement 2

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1. Self Rating

Area	Rating
LLM	C
Deep Learning	C
AI	C
ML	C

Explanation:
I am currently learning AI/ML concepts and actively building practical knowledge through coursework, certifications, and hands-on projects. I understand core concepts and can implement solutions with guidance. I am continuously improving my skills toward independently designing and developing AI/ML solutions.

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2. What are the key architectural components to create a chatbot based on LLM? Please explain the approach on a high-level

Key Architectural Components of an LLM-Based Chatbot (High-Level Approach)

1. Introduction:
AI: Artificial intelligence (AI) refers to a computer's ability to imitate or duplicate human cognition. It recognizes patterns and logics from given data that feed on to execute given tasks. For this AI use probability and predictions.

LLMs(Large Language Models): they are ai systems trained by using huge quantity of data(text, chats), so AI can analyze and understand trends in data. this an AI can generate return text(response) in chats. They are built using advanced neural network architectures. This is how most GPT models work.

Chatbot: these are computer programmes designed for interacting with humans by speech or text format conversation. this has certain good uses as well malicious uses.

good uses such as personal digital assistants, customer support bots, learning partner. malicious uses as spreading spam. mainly there 2 types of chatbots: Rule-Based Chatbots and artificial intelligence (AI) chatbots.

Rule Based Chatbots: these chatbots run on a fixed set of data, their interaction is simple. This type of chatbots, much used in customer services like Airtel, SBI, provide fixed options and answers in the same format.

Artificial Intelligence (AI) chatbots: These are the chatbots the we use in our daily life. This chatbots use Machine Learning(LLMs) to learn as they communicate with human. good examples for this are Chat GPT, Gemini etc.

2. Architecture:

It talks about the design or structure of how an AI model is built and how information flows inside it. We tell how AI thinks computationally by its architecture.

Diagram:

User → Frontend → Backend API → LLM → Vector DB → Response → User

1. User Interface (UI): this is the place where user interaction takes place.
   * Accept user input in text/voice/image
   * Display responses
   * Maintain chat experience
   * no intelligence involved here.
   * it developed using react.

2. Backend / Application Server: it acts the central controller and connects everything together.
   * Receives user requests
   * Authenticates users and Calls LLM APIs
   * Connects databases & tools
   * Manages workflow
   * it developed using node.js/express, python(flask api), java spring boot.

3. Large Language Model (LLM): they are ai systems trained by using huge quantity of data(text, chats).
   * Transformer architecture
   * Self-attention mechanism
   * Token prediction
   * Understand language and Reason over prompts
   * Generate responses
   * Examples - GPT Models.

4. Prompt Engineering & Context Management: giving input in a way that you can get your output in perfect condition.

* Format prompts
* Add system instructions
* Maintain conversation history
* Control response behavior
*eg:

System Prompt:
"You are a helpful AI tutor."

User Message:
"What is AI?"

5. Knowledge Retrieval System (RAG): it helps LLM in accruing external data as they don't have real-time data.

* Search database/vector store and Retrieve relevant documents from them.
* Then inject into prompt and send them to LLM
* its major components are Embeddings model, Vector database, Semantic search.
* example FAISS, Pinecone.

6. Memory & Data Storage: this where we store all conversation and information for a long time.

* Current chat context, Session history - short term memory.
* User preferences and Past interactions are long term memory.
* storage of conversations depends on system design.
* storage database - MongoDB Atlas, Vector DB.

7. Safety, Monitoring & Feedback: This layer is responsible for safety and reliable AI behavior with clients.

* it performs content moderation and Toxicity filtering from the data.
* Abuse detection and Logging conversations.
* Performance monitoring.
* it collects user feedback to better itself.
* It uses analytics dashboards to analyse itself and prevents harmful or incorrect outputs.

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3. Please explain vector databases. If you were to select a vector database for a hypothetical problem (you may define the problem) which one will you choose, and why?

01. 1.Vector Databases:

Introduction:

A vector database is a type of database used to store and search data in the form of numerical vectors called embeddings. These embeddings represent the meaning of data such as text or images. Vector databases are mainly used in AI applications where searching based on meaning is required instead of exact keyword matching.

02. **Concept of Vector Databases:**

In AI systems, text or documents are converted into embeddings using an embedding model. Similar content will have similar vector representations. A vector database stores these vectors and helps in finding related information by comparing similarity between vectors.

03. **Working Principle:**

First, data is converted into embeddings. These embeddings are stored in the vector database along with metadata. When a user gives a query, it is also converted into a vector, and the database retrieves the closest matching vectors using similarity measures like cosine similarity. This enables semantic search.

2.Hypothetical Problem:

Assume we are developing a **college academic assistant chatbot** that answers student questions using syllabus documents, exam regulations, and academic notices. Students may ask the same question in different ways, so the system must understand meaning rather than exact words and retrieve relevant information quickly.

3.Your Choice

For this scenario, I would like choose **ChromaDB** as the vector database.

4.Reason:

It is easy to set up and beginner-friendly for student projects.

1. It works smoothly with Python and LLM frameworks
2. It can run locally without complex cloud configuration.
3. It is very suitable for small to medium datasets like college documents.
4. It is Open-source and cost-effective for academic or trainee-level applications.